Preliminary Phanerozoic polar wander paths for the North and South China blocks

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Polar wander paths for the North and South China blocks suggest that (1) both were parts of Gondwana in the Palaeozoic, (2) the North China block accreted to Siberia in the late Permian and (3) the South China block accreted to the North China block in the middle Triassic to the early Jurassic. Comparison of the polar wander path for the South China block with that for northern Eurasia suggests that relative motion of over 4,000 km has occurred between them.

CHINA, like the rest of Asia, is composed of a number of tectonic blocks or cratons, separated by fold belts which mark the sutures where they have been welded together. Palaeomagnetism can be used to elucidate the origin and accretionary history of the blocks and is therefore important in geological studies of Asia[1-6]. We have made a detailed palaeomagnetic study of the North and South China blocks (NCB and SCB) to determine their apparent polar wander (APW) paths.

Regional geology and sampling

Geologically, the NCB is separated from Siberia by the Central Asian fold belts in the north and from the SCB by the Qinling fold belt in the south. The basement of the NCB consists of Archaean and early and middle Proterozoic metamorphic rocks. With the Luliang orogeny (~1,800 Myr) the parts of the NCB were welded together to form a cratonic block. We collected samples from Shandong, Liaoning and Shanxi provinces and Jixian County (Fig.1) and found that almost all of the periods from the Tertiary to the middle Proterozoic are represented. We collected a total of 1,981 samples from 250 sites but report here only the Phanerozoic results.

The SCB (the Yangtze paraplatform) is surrounded by fold belts, such as the early Palaeozoic South China fold belt to the south east and the Quinling fold belt to the north. It is connected to the NCB by the Huaiyang Shield in the north east. Most of the basement of the SCB is late Proterozoic. With the Jinning orogeny (~850 Myr) the SCB completed its transition from fold belt to cratonic block. We collected samples from Zhejing, Guizhou, Yunnan and Hubei provinces (Fig. 1) representing all the periods from the Tertiary to the late Proterozoic, the age of the oldest unmetamorphosed rocks in the SCB (1,029 samples from 180 sites).

Our samples were measured on cryogenic and spinner magnetometers using alternating field and thermal demagnetization methods. As many sites were demonstrably remagnetized, Zijderveld diagrams, vector differences and remagnetization circle methods were also used. All the results from this study and the few previous results satisfying the minimum criteria of reliability[4,5] are listed in the China palaeomagnetic data list[7], including 69 NCB and 90 SCB poles. The pole positions from our study are reported from sites, each consisting of five independently-oriented samples (data list available from the authors). The APW paths for the SCB and NCB (Fig. 2*a*, *b*) are constructed from the data in Table 1.

Palaeopoles from the SCB

Our Tertiary poles consist of determinations from three late Miocene basaltic lavas in Zhejiang Province (South China (SC) 1-3) giving a mean pole (SC 4). The results include antipodal normal and reverse flows, but with so few flows we cannot estimate how well secular variation has been averaged. The results, however, are consistent with studies of red beds in Hunan Province[3], which also pass a reversal test. Thus, whereas the Tertiary APW path clearly is not well constrained, it may include a loop towards northern Europe.

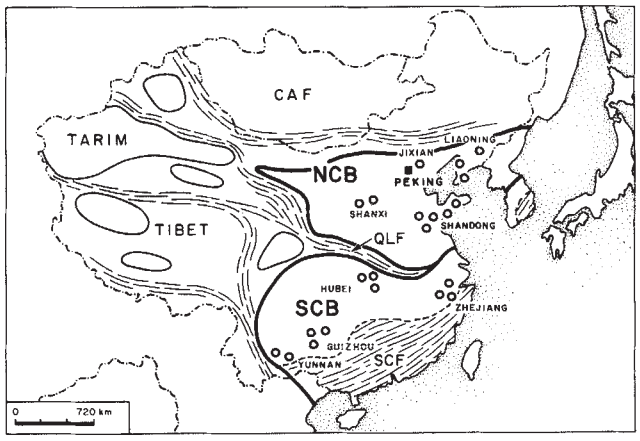


Fig. 1 Site map with major tectonic units in China shown schematically. The solid lines are block boundaries; CAF, Central Asian fold belts; QLF, Qinling fold belt; SCF, South China fold belt.

The Cretaceous results are from widely separated parts of the SCB; the Upper Cretaceous pole (SC 7) is for South Korea[8]. Our seven Lower Cretaceous sites in volcanic and sedimentary rocks are in Zhejiang, giving the mean pole SC17, consistent with SC 8 (ref. 9) and SC 9 (S. Sasajima, personal communication).

The mean Upper Jurassic pole (SC 24) from this study is based on six poles, SC 18-23, in Zhejiang ignimbrites and sandstones. The Middle Jurassic result (SC 27-30) from red beds in Guizhou province includes normally and reversely magnetized samples. Results from Hong Kong (SC 25) (S. Sasajima, personal communication) and Anhui province (SC 26) (ref. 10) are also included.

The Middle Triassic pole SC 32 (Chan Lung-san, personal communication) is preliminary, being based on eight samples from two sites, but the results do include two polarities and pass a fold test.

A collection of 42 samples from 13 sites in the Changxing carbonate section, a Permian/Triassic type section, yielded a consistent direction after thermal demagnetization to 300 °C. The resultant mean pole is SC 46, close to that obtained for the Middle Triassic.

Most of the 56 Permian samples studied from 10 sites in the Emeishan basalts of Guizhou carried a late Mesozoic or recent characteristic direction. Their magnetization was complex with at least four directions. Three of these are clearly secondary,

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Table 1 Mean palaeopoles for the North and South China Blocks

Age	$P_{lat.}$	$P_{long.}$	K	A_{95}	DP	DM	N	Test	China List
				SCB					
TM-TP	74.2	36.5	148	7.6			8	R	SC 1-5
KU	69.0	191.1	85	8.3	10.1	12.9	5		SC 7
KL	76.2	225.7		4.8			14	R	SC 8-17
JU	73.0	213.7	29	12.6			6		SC 18-24
JM	68.8	202.1		14.0			9	R	SC 25-30
TRM	54.6	209.7	95	(5.7)	3.3	6.2	2	RF	SC 32
PU-TRL	50.7	204.8	38	6.8	5.4	8.6	13		SC 33-46
PU	29.3	235.3	41	(12.0)	7.4	13.4	1	R	SC 47
CU	21.5	224.6	28	17.8	8.9	17.8	4	RF	SC 54-59
CL	25.2	220.7	12	(28.2)	14.3	28.4	1		SC 61
CBL	3.4	195.0	28	8.8			8	RA	SC 64-72,78
				NCB					
TM	76.4	178.0	230	6.0	7.2	9.3	4	R	NC 1-5
KU	69.0	182.0		4.0			8		NC 6
KL	69.0	200.0	40	12.2	12.5	17.5	5	R	NC 7-12
JU	71.3	225.7	38	7.9	6.7	10.3	10		NC 13-23
JU	74.2	215.6	57	4.0	3.7	3.4	10		NC 24
JM-JU	72.1	202.0	194	6.6	6.8	9.5	4		NC 25-29
JM	81.9	253.3	41	9.6	8.4	12.6	4		NC 30
TRL	35.3	39.3		17.6	14.0	22.0	6	R	NC 35-39
PU	38.1	6.3					2	R	NC 40-42
PU	44.0	358.0	40	6.9	3.8	7.2	12	R	NC 43
OM	43.2	332.5	53	(10.6)	5.3	10.6	1	R	NC 44
CBL	15.0	298.6	38	9.9			7	R	NC 46-60
PCU	16.5	301.1	35	9.5	8.8	12.9	8	R	NC 61-69

The database and statistics of the APW paths for the NCB and SCB. Column 1, age; columns 2 and 3, pole position; columns 4 (K) and 5 (A_{95}), precision parameter and 95% error in Fisher's[39] analyses; columns 6 (DP) and 7 (DM), semi-axes of the elliptical error around the pole at 95% probability, DP in the colatitude direction and DM perpendicular to it; column 8 (N), number of sites; column 9, stability tests, R for reversal test, F for fold test and A for attitude test; column 10, pole number in China palaeomagnetic data list[7]. Abbreviations: TM-TP, Teritary Miocene-Pliocene; KU, Upper Cretaceous; KL, Lower Cretaceous; JU, Upper Jurassic; JM, Middle Jurassic; TRM, Middle Triassic; PU-TRL, Upper Permian-Lower Triassic; CU, Upper Carboniferous; CL, Lower Carboniferous; CBL, Lower Cambrian; OM, Middle Ordovician; PCU, Upper Precambrian.

being in the present dipole field direction and in the Jurassic normal and reversed directions. We tentatively identify the fourth as a primary Permian direction which gives the pole SC 47, approximately antipodal to the Permian/Triassic boundary result. A recently-reported Permian pole for the SCB[6] is inconsistent with our result. It may be a secondary remagnetization[7], or the result of tectonic rotation (M. W. McElhinny, personal communication), so is not included in the APW path described here.

The Upper Carboniferous poles (SC 54-59) were obtained from reversely magnetized samples, consistent with a Kiaman reversed interval age. The samples were from a fresh man-made outcrop and were not as severely remagnetized as the remaining Carboniferous sites. The Lower Carboniferous pole (SC 61) was obtained from normally magnetized results. Thus, our Carboniferous results are based on data which pass a reversal test. They also pass a fold test.

We did not obtain useful data from Devonian, Silurian or Ordovician sites, all of which were remagnetized strongly in Recent or Mesozoic directions.

The Lower Cambrian poles are based on collections from Zhejiang, Hubei and Yunnan provinces. The poles from the Hetang Formation in Zhejiang and the Tianheban Formation almost 1,000-km away in Hubei satisfy an attitude test at the 95% confidence level but are discrepant before bedding correction. The Tianheban is flat lying. After the Hetang Formation results have been corrected for early Palaeozoic tilt, the data from both formations are in good agreement, this test being comparable with a fold test.

Palaeopoles from the North China block

The Tertiary poles (North China (NC) 1-5) include both normal and reversed polarities, but are preliminary because they are both from the same area (Linju County of Shadong Province) and from basalts. The Upper Cretaceous pole (NC 6; ref. 11) and the Lower Cretaceous poles (NC7-12) are from andesites and rhyolites in Zhuji County of Shandong. The Upper Jurassic sites (NC 13-23) are in sandstone of the Laiyang Formation in Zhuji County of Shandong, consistent with (NC 24; ref. 10), obtained from the Mengying Formation in western Shandong. The Middle/Upper Jurassic poles (NC 25-29) are for siltstones from quarry sites of the Santai Formation in Zibo of Shandong. In all these studies, alternating field and thermal demagnetization methods gave consistent results.

Preliminary Lower Triassic results from red beds in three areas have similar secondary magnetizations. (The results from Zibo, Shandong Province are so close to the present field direction before bedding correction that they are not included.) Two sites from Taiyuan of Shanxi Province give scattered results with a northwesterly declination and shallow positive inclination. The four sites from Benxi of Liaoning Province include both normal and reversely-magnetized rocks. The normal samples have a similar direction to those from Shanxi Province. The direction in reversely-magnetized samples migrates towards that antipodal to the normal samples on thermal demagnetization, but even at high temperature fail to reach the antipodal direction. The poles from Liaoning Province are included (NC 35-39).

The Upper Permian poles (NC 40-42) and (NC 43; ref. 6) are from a Permian/Triassic sedimentary section in Taiyuan of Shanxi. Sandstones in the section are heavily remagnetized, but mudstones gave consistent reversed directions after thermal demagnetization.

Most Ordovician samples were remagnetized so strongly that no useful information could be obtained; the Middle Ordovician pole position reported here (NC 44) is from only one site but is based on AF and thermal demagnetization, hence we tentatively suggest that it is an Ordovician characteristic direction.

The Lower Cambrian pole positions (NC 46-56) and (NC 58-60) from oolitic carbonate in Zhangxia of Shandong and from carbonate in Fuzhouwan of Liaoning have different thermally-cleaned directions from any observed Phanerozoic

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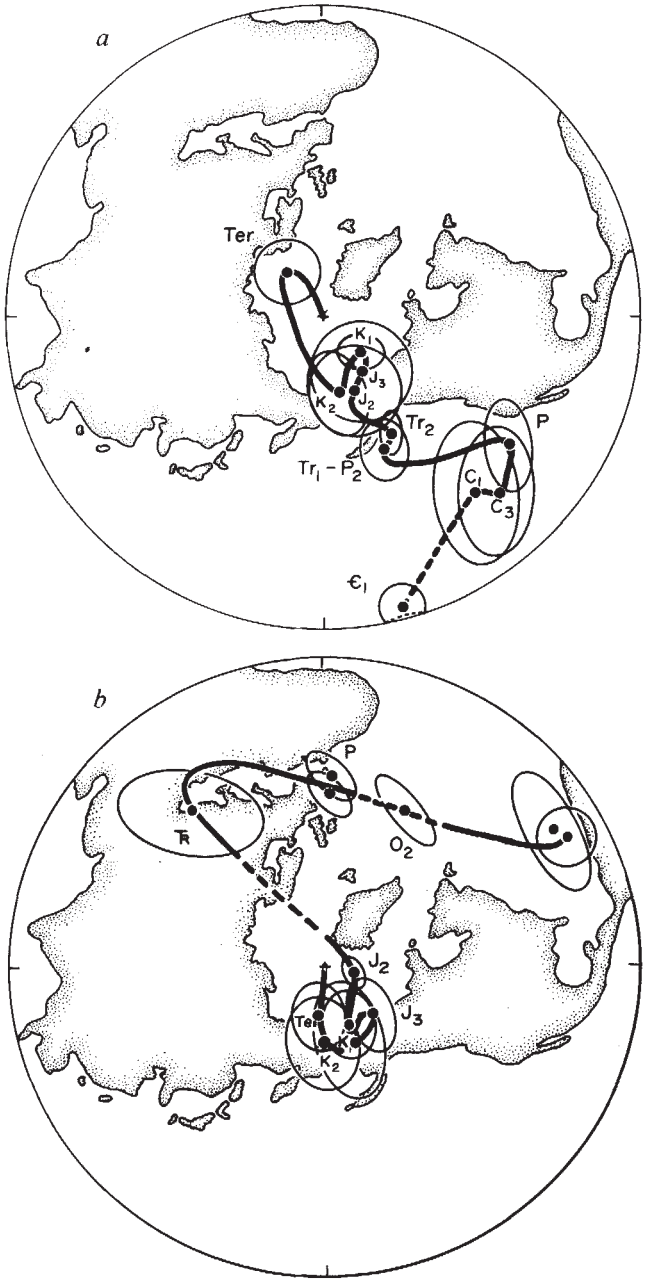


Fig. 2 *a*, APW paths for the SCB; *b*, the NCB. Ter, Tertiary; K_2, Upper Cretaceous; K_1, Lower Cretaceous; J_3, Upper Jurassic; J_2, Middle Jurassic; Tr_2, Middle Triassic; Tr_1, Lower Triassic; P_2-Tr_1, Upper Permian to Lower Triassic; P, Permian; C_3, Upper Carboniferous; C_1, Lower Carboniferous; O_2, Middle Ordovician; E, Canbrian; E_1, Lower Cambrian. Circles and ellipses are the A_{95} or α_{95} confidence intervals of pole position. The poles used in construction of the two APW paths are indicated in Table 1, where the relevant statistics are also given.

characteristic directions. Lower Cambrian sandstones in North Korea have a similar direction (NC 57; ref. 11). We therefore suggest that this is a Lower Cambrian characteristic direction. The pole positions from the Upper Precambrian Nanfen Formation (NC 61-69) are very close to the Lower Cambrian pole position, supporting the suggestion that the Nanfen Formation is close to early Cambrian in age.

Summary of palaeomagnetic results

The APW paths can be traced back unambiguously to the Carboniferous for the SCB and to the Permian for the NCB but the middle Palaeozoic poles require much more work on both blocks. In contrast, the Cambrian poles are well established for

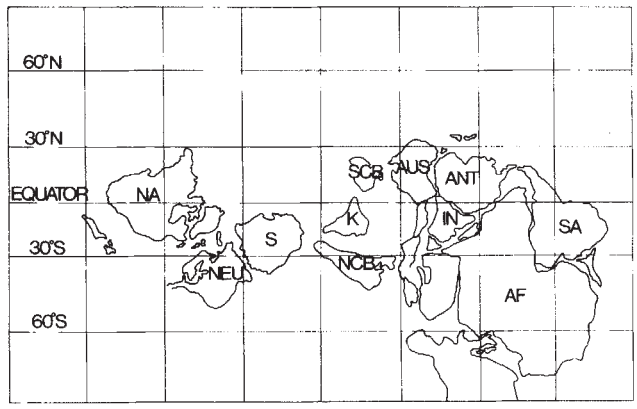


Fig. 3 Lower to Middle Cambrian reconstruction modified from ref. 13. AF, Africa; ANT, Antarctica; AUS, Australia; IN, India; K, Kazakhstan; NA, North America; NEU, northern Europe; S, Siberia; SA, South America. Note that the NCB and SCB are treated as separate blocks.

each block. Because of the absence of middle Palaeozoic data, there is a polarity ambiguity which is resolved by assuming that the polar wander between the age of the oldest reliable late Palaeozoic results and the Cambrian was less than 90°.

When compiling the APW paths for the major continental blocks, palaeopoles have been selected according to four criteria: demagnetization, statistics, the accuracy of ages and possible tectonic rotations[12]. All our results (Table 1) were based on either thermal or alternating field demagnetization, or both methods. With the exception of SC 061, which has a 95% confidence radius of 28.2°, all the results in Table 1 are based on palaeopoles having confidence intervals <25°. Most of our samples were collected from stratigraphic type sections for which there is good fossil control. The ages of the Phanerozoic units sampled are usually elucidated at stage level and at least to the series (that is, Lower, Middle and Upper) level in terms of the geological time-rock units, so that their ages are generally known to better than 25 Myr. As we sought to obtain APW paths for the cratonic NCB and SCB, we collected neither from fold belts nor from tectonically-active areas near the margins of the two blocks. These data therefore satisfy standard palaeomagnetic criteria (for example, ref. 12).

In addition, reversal and fold or attitude tests are satisfied by some of the sites (Table 1). APW paths for the NCB and the SCB therefore should be sufficiently reliable to begin to test tectonic hypotheses, although they will be refined by later data.

Cambrian palaeogeography

Based on the palaeomagnetic results, we propose that: (1) the NCB and SCB were separate in the Cambrian because their palaeopoles are quite different; (2) the SCB is placed adjacent to northern Australia, occupying an equatorial latitude, whereas the NCB is placed near to Tibet, northern India and Iran at 35° latitude in the Southern Hemisphere (Fig. 3). Thus both the NCB and the SCB were parts of Gondwana, but were separated from eath other.

The proposed position of the SCB has the following merits: (1) the strong affinity of the Cambrian trilobites between the SCB and Australia is explained readily[15-17]; (2) the sedimentary facies belts in the SCB and Australia match each other, for example, the sequence from the west to the east is, first, a clastic belt, then the miogeosynclinal facies belt of shallow marine carbonate and finally, the eugeosynclinal facies belt of deep-marine cherts or flysch facies[14,17,18]; (3) the huge phosphorite deposits in Yunnan Province and in North Vietnam of early Cambrian age and those of middle Cambrian age in Hainan Island of the SCB and the Georgina Basin of Australia would all have been formed adjacent to each other in Cambrian tropical marine basins; (4) archaeocyathids in the early Cambrian and gypsum and salt pseudomorphs in middle and late Cambrian

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of the SCB and Australia are consistent with the tropical and low latitudes derived from the palaeomagnetic results.

The Southern Hemisphere position for the NCB depends on interpretation of the polarity of its Cambrian pole. It is supported by the distributions of the Cambrian trilobites[15] and early Ordovician cephalopods[19], as well as the well-documented Cambrian evaporite belt from the Arabian Peninsula via Iran, Pakistan, northern India to North-west and North China[20–22]. This all suggests that in the early Palaeozoic the North China block was close to Tibet, northern India and Iran (see ref. 7).

Permian reconstruction

The late Permian palaeomagnetic results from the NCB show that it was then very close to the equator; the Permian reconstruction described here, with the Chinese blocks in Tethys, next to northeast Africa and the Arabian Peninsula (Fig. 4), is different from previous reconstructions[13,23,24].

The merits of this new reconstruction are: (1) it satisfies the low palaeolatitudes of the Chinese blocks suggested by Permian fossils; (2) it explains the Permian flora distribution, for example, the four flora provinces of Angara, Euramerica, Gondwana and Cathaysia are all grouped together and at the appropriate latitude in this reconstruction; (3) it is consistent with the tectonic history of the region, for example, the Central Asian fold belt was sutured finally in the late Permian by a process which started in the west and propagated eastwards. This is the so-called 'migration of geosyncline'[27], which we interpret as the collision of the NCB with the Siberian block starting from Northwest China and extending to Northeast China (see ref. 7).

Assembly of the China blocks

The Cambrian poles from Shandong and Liaoning provinces of North China are different from the Zhejiang, Yunnan and Hubei provinces of South China (Table 1), demonstrating that the NCB and SCB were distant from each other. In contrast, the Middle Jurassic pole from Shandong Province of the NCB and those from Zhejiang Province of SE China and Guizhou Province of SW China are statistically indistinguishable, implying that the NCB and SCB were assembled as a single unit by the middle Jurassic[7].

The Permian results from Shanxi Province show that the NCB had an equatorial latitude in the late Permian. Similarly, the late Permian basalt direction in Guizhou Province shows that the SCB was also in an equatorial region during that period. The relative orientations of the two blocks, however, were quite different in the Permian from those at present, suggesting that although they were close to each other in the Permian, they were not welded into a single tectonic unit. The palaeomagnetic results are consistent with the evidence from palaeontology and sedimentary facies[28,29]. We have no detailed palaeomagnetic results from Middle Triassic to Lower Jurassic for either blocks, preventing accurate timing of the amalgamation of the NCB and SCB; analyses of sedimentary facies in the two blocks[28] suggests that the amalgamation occurred during the Triassic Indosinian orogeny.

Eastward movement of the SCB

The palaeomagnetic pole positions from the NCB and SCB are displaced to the east of the northern Eurasia path (Fig. 5*a*). Nevertheless, the Mesozoic segments of the APW paths for the SCB and northern Eurasia are similar in form. Rotation of the former clockwise by 50° about an Euler pole near the present north pole brings the two paths into agreement (Fig. 5*b*). Such a rotation brings the SCB westward to the region presently occupied by western India and Pakistan, suggesting that relative motion of about 4,000 km occurred between the SCB and northern Eurasia. The SCB is thought to have been pushed eastward during the Cenozoic when the Indian plate collided with the Eurasian plate in the Eocene and subsequently as it continued its northward motion[30,31]. Our palaeomagnetic data suggest a far larger displacement than the proposed eastward motion, which is only several hundred kilometres.

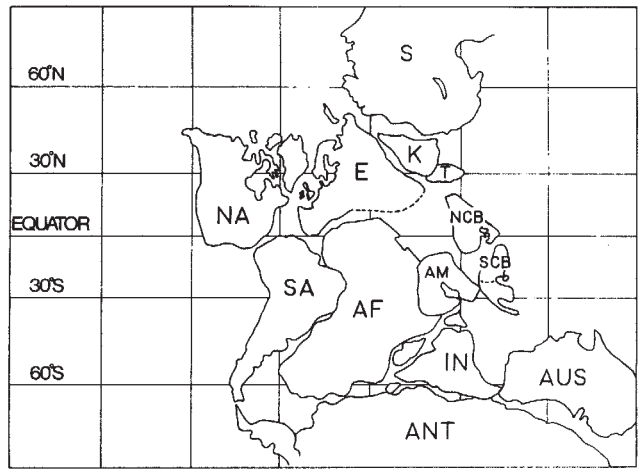


Fig. 4 Permian reconstruction. AM, Asia Minor; E, Europe; I, India; T, Tarim; remaining abbreviations as Fig. 3. The grouping of Gondwana and Laurasia is adopted from the Pangaea A[25,26]. The relative configuration between Gondwana and Laurasia and the positions of the Chinese blocks are based on the Pangaea B[26]. Note that although the Tarim block may have been accreted already to the Kazakhstan block so that they share the same Angara flora, the NCB and SCB were still distant from the Siberian block. The accretion of the NCB to the Siberian occurred in the late Permian.

We sought to establish whether the difference between the APW paths of South China and northern Eurasia is statistically significant. We consider the segments of the two APW paths from the Carboniferous to late Cretaceous. The APW path segment for the SCB consists of nine poles, that is poles of K2, K1, J3, J2, TR2, P2-TR1, P2, C3 and C1. They all fall to the east of the corresponding poles from northern Eurasia, which is highly improbable (1 in 512) if the two groups of poles are statistically indistinguishable. The α_{95} confidence intervals for the palaeopoles from both northern Eurasia and South China are for the most part $<15°$. Using these values, there is overlap between the confidence intervals of some of the Mesozoic poles of the two paths. Using the standard errors ($P = 0.63$) of other palaeomagnetic workers when comparing APW paths[12], however, there is no overlap. The two curves are therefore statistically distinguishable.

The inclination data and the implied palaeolatitudes of the SCB show that the observed difference in the northern Eurasia and SCB APW paths does not result from a local rotation (Fig. 6). Assuming no relative motion between the SCB and northern Eurasia, the palaeolatitude is significantly lower at Hongzhou than that from the northern Eurasia APW path. Thus the difference in APW paths cannot be owing to a local rotation, but must involve substantial relative motion between the two regions. The palaeolatitudes become statistically indistinguishable for westward rotation of the SCB by 45, 50 or 55° about an Euler pole close to the north pole with a best fit of 45° for the last 75 Myr, whereas a rotation of more than 55° is needed to fit the segments before 225 Myr. Therefore, there may have been two stages of relative movement between South China and northern Eurasia.

The amount of relative motion along the boundary between two blocks is very dependent on the distance from the Euler pole, reaching a maximum when the pole is 90° from the boundary. Fitting the Mesozoic path segments gives an Euler pole at 86.5N31.5E with angle of rotation 50° clockwise and a displacement of 4,300 ± 1,200 km eastward for the SCB path with respect to the northern Eurasia path.

We conclude from the analysis of the APW paths that the SCB has moved eastward with respect to northern Eurasia by ~4,000 km. Two fundamental questions remain: first, as the SCB and Siberia are separated presently by 2,500 km, how were the 4,000 km of relative motion distributed between the intervening tectonic units? Second, when did this motion take place?

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The Middle Jurassic poles from the NCB and SCB are statistically indistinguishable, differential motion towards this pole cannot therefore have been more than 1,500 km since the middle Jurassic, using the 63% errors of these poles. This suggests that a major portion of the relative motion between the SCB and northern Eurasia occurred along the Central Asian fold belt forming the boundary between the NCB and northern Eurasia. This fold belt is the largest in the world; it is aligned approximately east-west and is still active[32,33]. It has been considered already as a possible plate boundary between the seismically-active Chinese blocks and the aseismic northern Eurasia plate[34]. We consider that much of the relative motion, required by the APW paths to have occurred between the SCB and northern Eurasia, may have been along this Central Asian fold belt.

It is tempting to assign the relative motion to the Cenozoic and to interpret it as a consequence of the Indian-Eurasian continental collision. First, a single rotation brings the two Mesozoic APW paths into agreement. Second, the relative motion suggested by the palaeomagnetic data is just equal to the width of the northern India (3,000 km) plus the width of the transverse ranges between India and the SCB (1,000 km), consistent with a rigid indentor displacing parts of the southern boundary of Asia along previously established lines of weakness. Our data, however, do not preclude motion between the SCB and northern Eurasia during the Mesozoic because of the inherent limits of the resolution of the palaeomagnetic method. We do know that the NCB rotated substantially with respect to both Eurasia and the SCB during the Triassic. Our interpretation is that relative motion of at least 2,500 km occurred between the NCB and northern Eurasia during Cenozoic, possibly along the Central Asian fold belt in addition to the southern belts[30,31].

Accretionary history of China

Our study has revealed a complicated geological history for the NCB and SCB. In the Cambrian, both blocks seem to have been part of Gondwana, but were separated from each other. The SCB had an equatorial palaeolatitude and was near northern Australia whereas the NCB was at 35° S, close to Iran, Tibet and northern India. During the period from the late Ordovician to Carboniferous, Gondwana moved across the South Pole, causing an inversion of the latitude zones[13,35]. The NCB then led the SCB in their common northward movement. In the late Carboniferous and early Permian, the NCB arrived near the palaeoequator; in the Permian, the NCB was in Tethys, next to the Iranian, Turkish and Arabian blocks. The SCB had become separated from Australia by the Devonian and moved toward the NCB. In the late Carboniferous and early Permian, both blocks were in Tethys with the Indochina block. During the late Permian, the NCB started to move northward and collided with Siberia along the Central Asian fold belt. It was not until the middle Jurassic, however, that the NCB and SCB were welded together. Since the beginning of the Cenozoic, when the Indian plate moved northward and collided with the Eurasian plate, the Chinese blocks have been pushed further eastward. In particular, the SCB has moved eastward with respect to northern Eurasia.

The following sequence is suggested: (1) Kazakhstan accreted to Siberia at the end of the Carboniferous[36]; (2) Russia collided with Siberia in the early Permian[37]; (3) the NCB accreted to Siberia in the late Permian[7]; (4) the SCB collided with the NCB by the middle Jurassic, possibly during the late Triassic; (5) central Iran accreted to Eurasia in the early Cretaceous; (6) India collided with Eurasia during the Eocene[30]; (7) the Arabian Peninsula separated from Africa and has moved toward Asia since the late Tertiary (15 Myr); (8) 50 Myr in the future, Australia may collide with Eurasia. It thus seems that Asia has grown by accreting fragments from Gondwana. Conversely, these fragments have been drifting away from Gondwana for the past 250 Myr. This remarkably long history of rifting around Africa may result from insulation[38] beneath the largely stationary core of Gondwana, which produces the necessary heat accumulation to drive the rift systems.

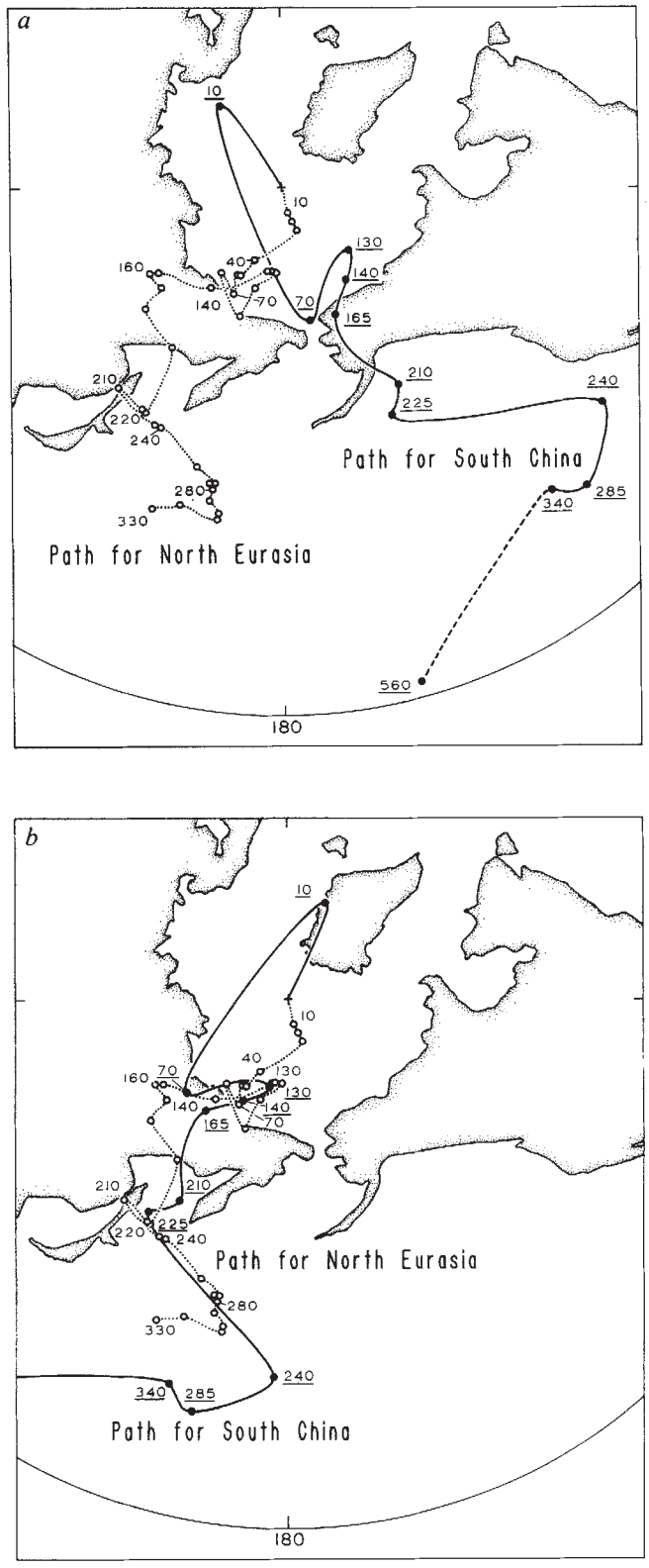


Fig. 5 Comparison of the APW paths for northern Eurasia[12] (dotted line), and for the SCB (solid line). Open circles, pole positions at 10 Myr intervals from northern Eurasia; solid circles, those from the SCB; numbers, suggested ages of the poles (Myr). The ages for the poles of the SCB are underlined. *a*, The South China APW path is displaced to the east with respect to the northern Eurasia path. *b*, A 50° clockwise rotation about an Euler pole near the present north pole brings the Mesozoic and late Palaeozoic segment of the Chinese path into agreement with that of the northern Eurasia path. The same rotation brings the SCB westward to a position presently occupied by western India and Pakistan.

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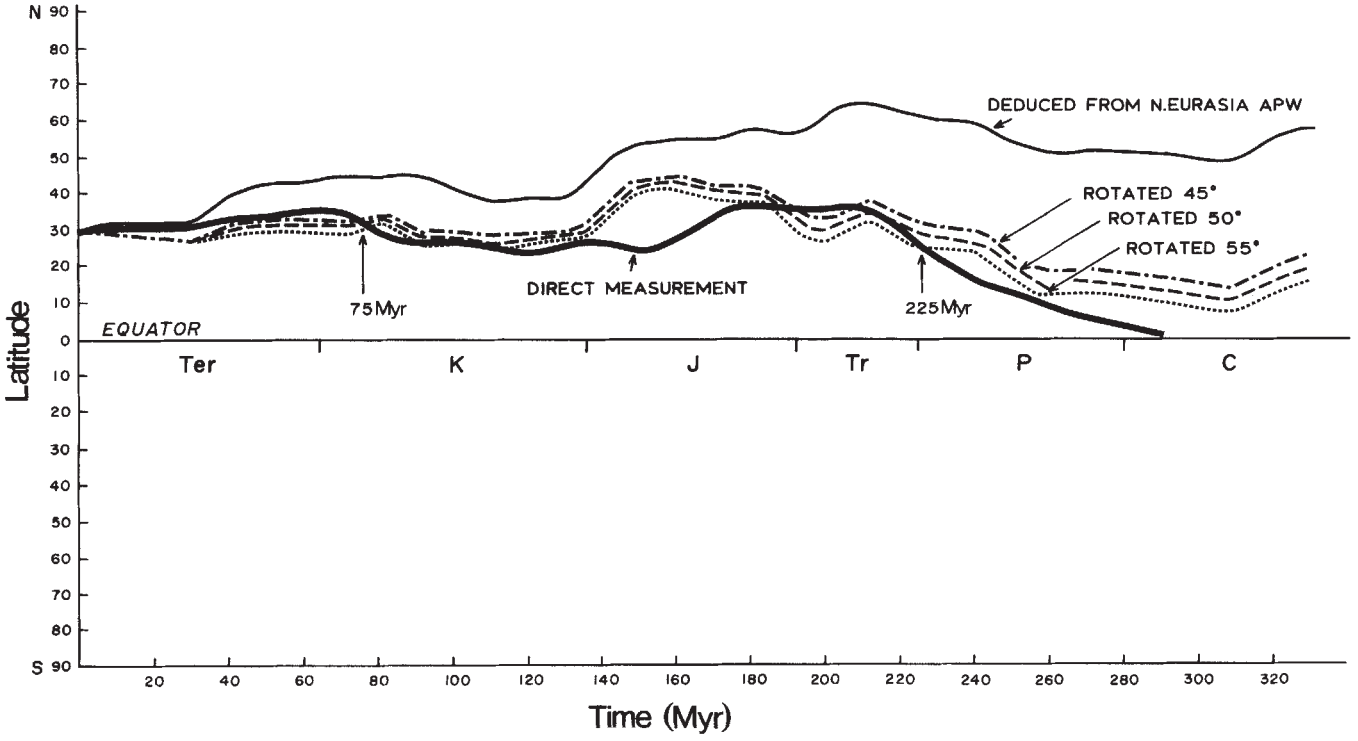


Fig. 6 Palaeolatitudes of the Hongzhou in the SCB determined from the South China samples (heavy solid curve) and those deduced from the northern Eurasia APW path (remaining curves). The uppermost curve shows the palaeolatitudes deduced from the northern Eurasia APW path with the SCB at its present position; the rest are those deduced with the SCB rotated westward by 45, 50 and 55°, respectively.

Conclusions

Preliminary APW paths for the NCB and SCB provide the palaeomagnetic constraints for these blocks in palaeogeographic reconstructions; the palaeomagnetic results from these blocks differ from those of Siberia and Europe and the Chinese APW paths are displaced systematically to the east of the northern Eurasia APW path. This eastward shift of the South China APW path suggests that the SCB has moved eastward 4,000 km with respect to Siberia, but the details of when and how this motion took place are as yet unknown.

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